

# Does Video Count Drive Creator Success?

YouTube Creator Analytics · UIUC MSBA · 2026

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75

Leading  
Creators

3

Content Categories

YouTube  
Data API v3

# Project Overview

## Common Assumption

*"The more frequently you upload,  
the faster your channel grows."*

### $H_0$ (Null):

Upload frequency has NO significant effect on subscriber count.

### $H_1$ (Alternative):

Upload frequency positively predicts creator success.

01

### Data Collection

YouTube Data API v3 · 75 leading creators collected via API search across 3 categories

02

### Data Preparation

Merged CSVs · Engineered subs\_per\_video & views\_per\_video

03

### Exploratory Analysis

Correlation matrix · Log-scale scatter · Category medians

04

### Statistical Testing

OLS Regression + Kruskal-Wallis non-parametric test

05

### Tableau Dashboard

3 interactive dashboards covering landscape, performance & trends

# Dataset & Sample Design



**25 Creators**

Entertainment



**25 Creators**

Education



**25 Creators**

Science &  
Technology

## Variables Collected

| Variable         | Type        | Description                                | Role               |
|------------------|-------------|--------------------------------------------|--------------------|
| subscriber_count | Continuous  | Total channel subscribers                  | Outcome Variable   |
| video_count      | Continuous  | Total videos published                     | Predictor Variable |
| view_count       | Continuous  | Total channel views                        | Supplementary      |
| category         | Categorical | Entertainment / Education / Science & Tech | Group Variable     |

# Statistical Results — OLS Regression

0.025

$R^2$

Explains only 2.5% of subscriber variance

0.156

Coefficient

Weak positive trend (more videos → slightly more subs)

0.175

p-value

NOT significant — fail to reject  $H_0$  ( $p > 0.05$ )

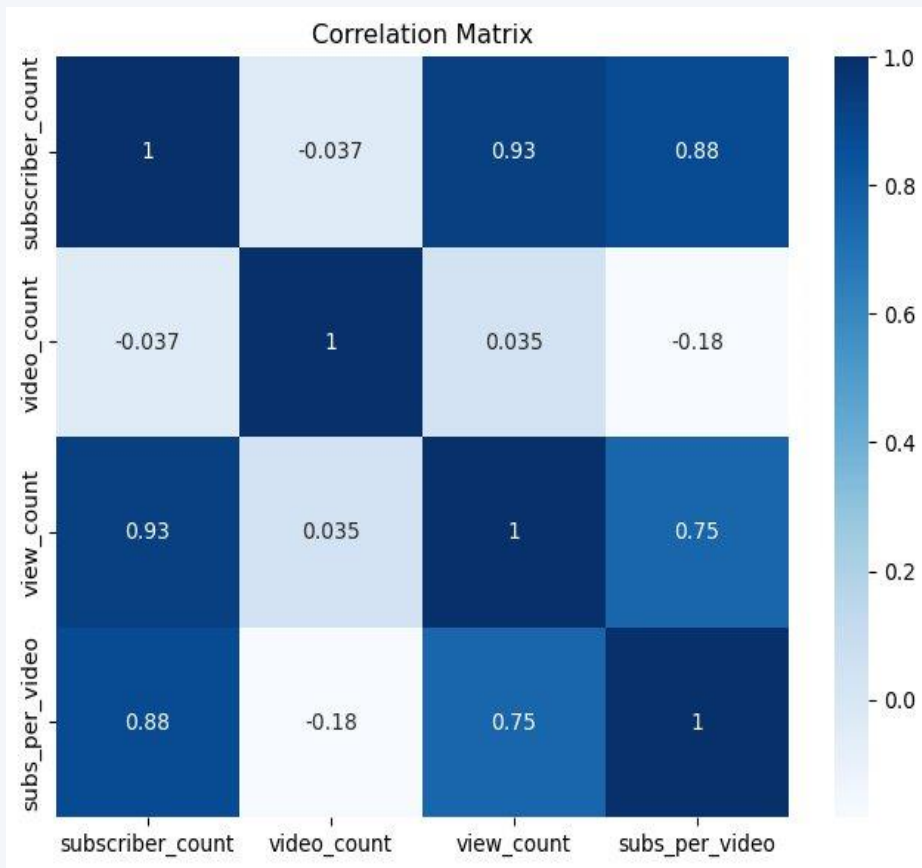
## OLS Regression Results

```
=====
Dep. Variable:    subscriber_count    R-squared:                0.025
Model:            OLS                Adj. R-squared:           0.012
Method:           Least Squares       F-statistic:              1.872
Date:             Mon, 08 Jun 2026    Prob (F-statistic):       0.175
Time:             11:50:33            Log-Likelihood:           -118.32
No. Observations: 75                 AIC:                     240.6
Df Residuals:     73                 BIC:                     245.3
Df Model:         1
Covariance Type:  nonrobust
=====
```

```
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
const         15.0130     0.811     18.510     0.000     13.396     16.629
video_count    0.1557     0.114      1.368     0.175     -0.071     0.383
=====
```

```
=====
Omnibus:                 3.669    Durbin-Watson:           0.901
Prob(Omnibus):            0.160    Jarque-Bera (JB):        4.008
Skew:                     0.017    Prob(JB):                0.135
Kurtosis:                 4.132    Cond. No.                43.0
=====
```

# Correlation Matrix



## Key Correlations

subscriber ↔ video\_count

**$r = -0.037$**

Near-zero — upload volume does NOT drive subs

subscriber ↔ view\_count

**$r = 0.93$**

Very strong — engagement drives channel size

subs\_per\_video ↔ video\_count

**$r = -0.18$**

More videos → slightly lower efficiency

# Category Comparison — Kruskal-Wallis Test

Kruskal-Wallis Test → Statistic: 28.74 | p-value:  $5.72 \times 10^{-7}$  | Result: Highly Significant ( $p < 0.001$ )

## Entertainment



Most Efficient

Avg. Subscribers

**44.5M**

Avg. Videos

**1,800**

Median Subs / Video

**27,979**

## Education



2nd

Avg. Subscribers

**14.7M**

Avg. Videos

**2,255**

Median Subs / Video

**11,274**

## Science & Tech



3rd

Avg. Subscribers

**6.9M**

Avg. Videos

**2,955**

Median Subs / Video

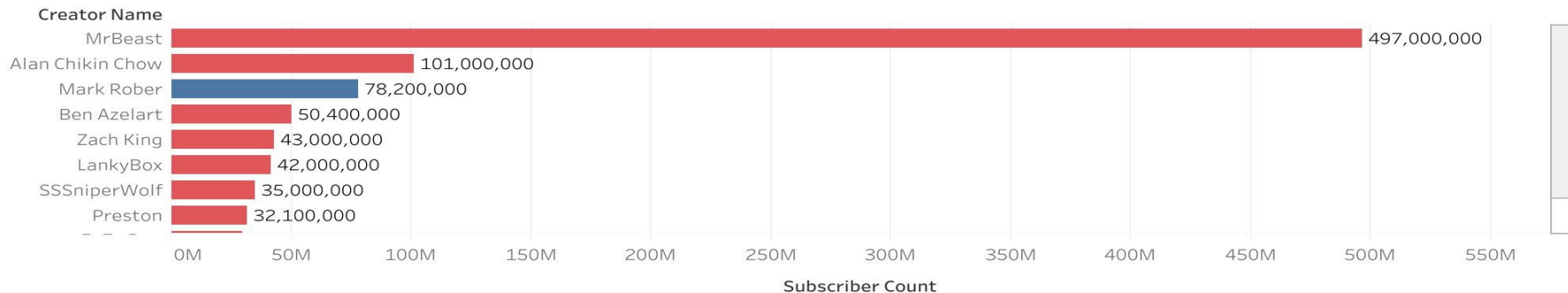
**3,120**

*Median subs/video used to reduce skew from outliers (MrBeast: 504K, Mark Rober: 303K subs/video).*

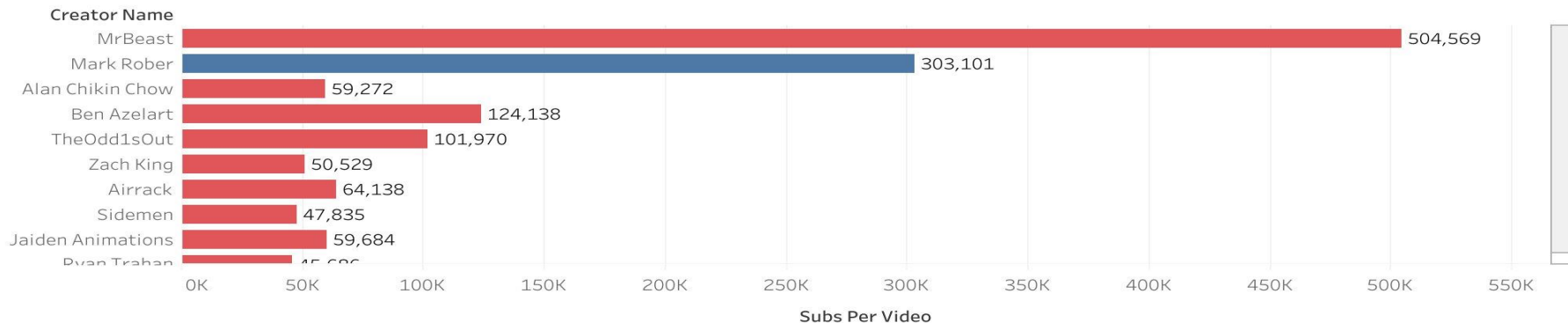
*Results remain consistent when using median efficiency, reducing the impact of outliers such as MrBeast and Mark Rober.*

# Creator Performance — Top Rankings

## Top 10 Creators by Subscribers



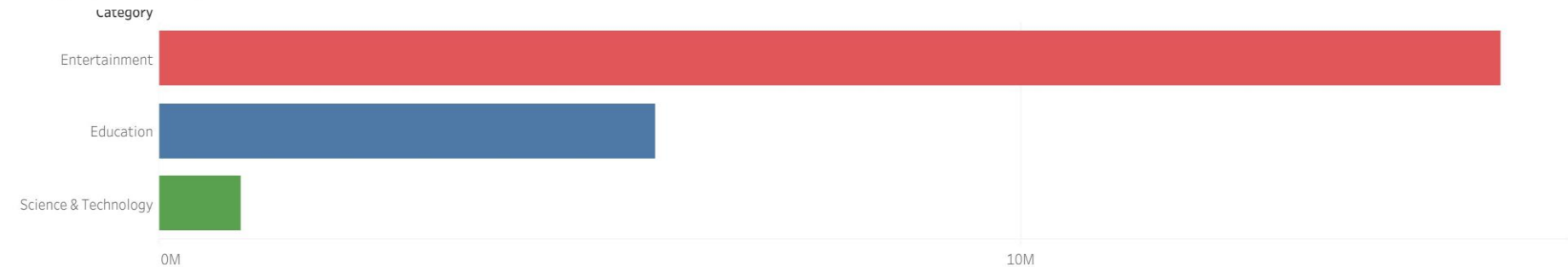
## Top 10 Most Efficient Creators



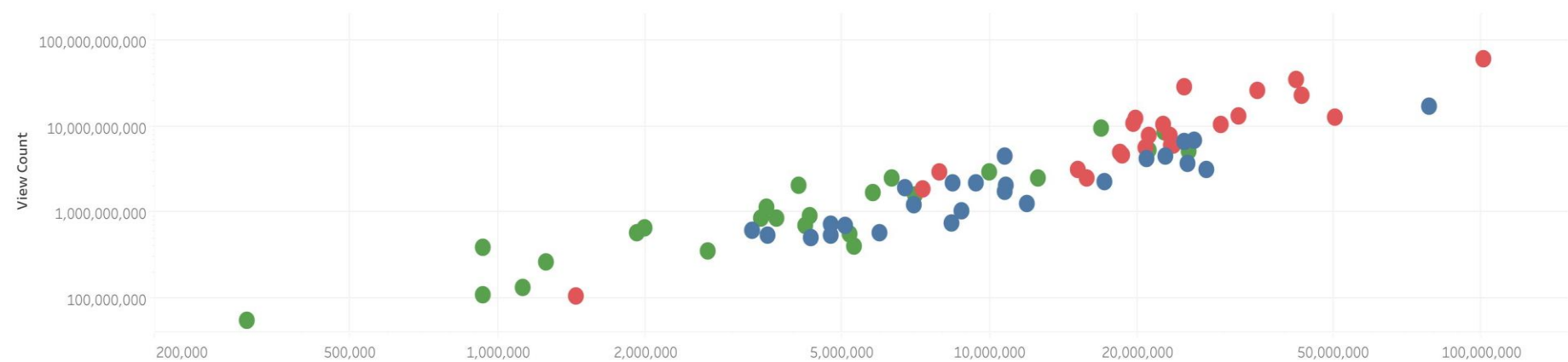
# YouTube Creator Landscape — Category Overview

Category All

## Category Summary



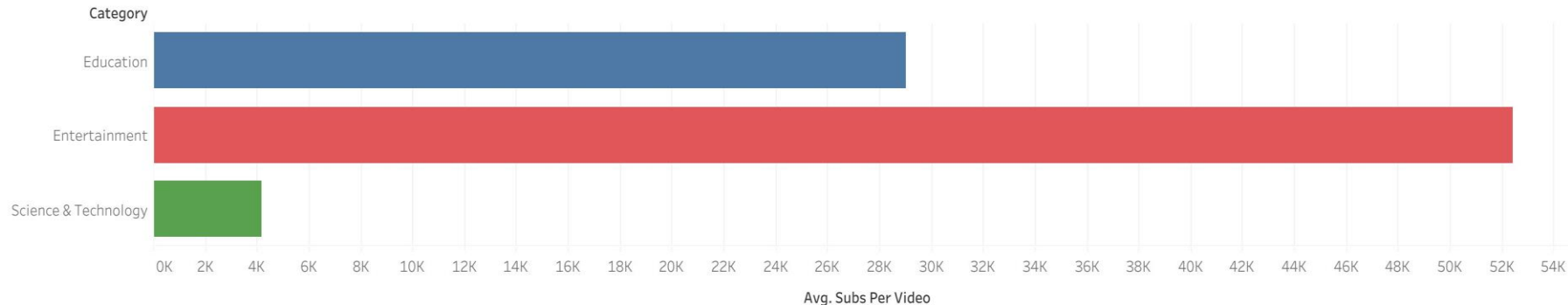
## Subscriber vs Views



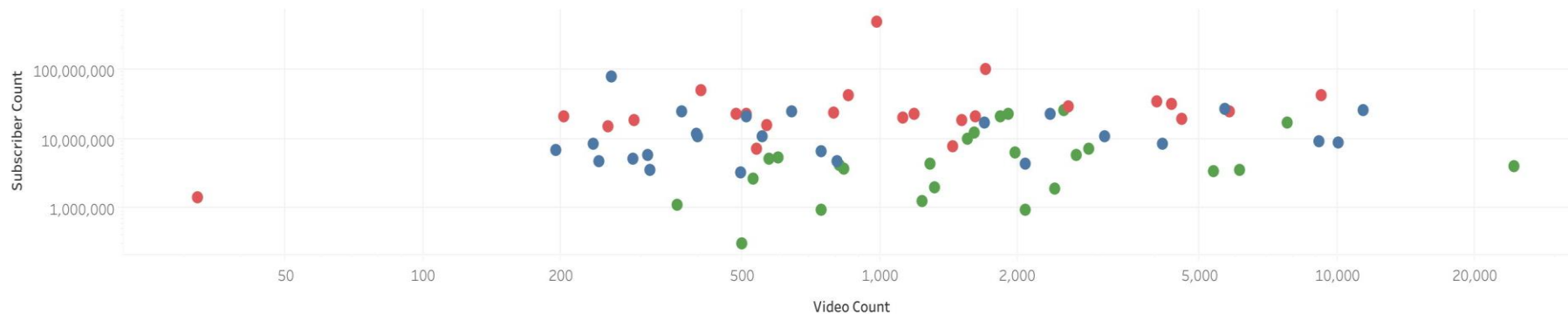


# Content Efficiency & Video Count vs. Subscribers

Average Content Efficiency by Category



Video Count vs Subscriber Count



# Key Findings

1

## Video Count Does NOT Drive Growth

$R^2 = 0.025$ ,  $p = 0.175$  — fail to reject  $H_0$ .  
Video count explains only 2.5% of variation in subscriber count.

2

## Near-Zero Correlation Confirmed

Correlation between video\_count & subscriber\_count is -0.037. Higher upload volume links to lower per-video efficiency ( $r = -0.18$ ).

3

## Engagement — Not Volume — Drives Growth

subscriber\_count  $\leftrightarrow$  view\_count  
correlation:  $r = 0.93$ . Audience engagement is the primary channel growth driver.

4

## Entertainment is 12.6× More Efficient

52,416 subs/video vs 4,150 for Science & Tech. Kruskal-Wallis confirms category differences are highly significant ( $p = 5.72 \times 10^{-7}$ ).

5

## Quality Outliers Beat High-Volume Creators

Mark Rober: 78M subs, only 258 videos  $\rightarrow$  303K subs/video. MrBeast, Kurzgesagt & TheOdd1sOut show the same pattern.

# Conclusion

*Upload frequency alone is NOT a meaningful predictor of YouTube creator success.*

- Video count explains only 2.5% of variation in subscriber count ( $R^2 = 0.025$ )
- The correlation between upload volume and subscribers is effectively zero ( $r = -0.037$ )
- Content category, engagement quality, and audience fit are far more important than posting frequency
- Entertainment creators achieve 12.6× higher subscriber efficiency than Science & Technology creators — with fewer videos

**Strategic Recommendation: Invest in content quality and audience alignment — not upload frequency.**